

# **SIMATS ENGINEERING**

## **CO1**

### **ASSESSMENT TOOL 2**

**COURSE NAME: CSA07**

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**COURSE CODE: Computer Networks**

**REG NO : 192521072**

#### **1. Given**

- **IP Address Block: 192.168.100.0/24**
- **School of Computing: /25**
- **School of Electronics: /26**
- **School of Mechanical Engineering: /27**

Subnet masks:

- **/25 = 255.255.255.128**
- **/26 = 255.255.255.192**
- **/27 = 255.255.255.224**

#### **Step 1: Allocate the Subnets**

##### **1. School of Computing (/25)**

- **Network Address: 192.168.100.0**
- **Subnet Mask: 255.255.255.128**
- **Broadcast Address: 192.168.100.127**
- **Valid Host Range: 192.168.100.1 – 192.168.100.126**
- **Usable Hosts: 126**

##### **2. School of Electronics (/26)**

- **Network Address: 192.168.100.128**
- **Subnet Mask: 255.255.255.192**
- **Broadcast Address: 192.168.100.191**
- **Valid Host Range: 192.168.100.129 – 192.168.100.190**
- **Usable Hosts: 62**

### 3. School of Mechanical Engineering (/27)

- **Network Address:** 192.168.100.192
- **Subnet Mask:** 255.255.255.224
- **Broadcast Address:** 192.168.100.223
- **Valid Host Range:** 192.168.100.193 – 192.168.100.222
- **Usable Hosts:** 30

### Step 2: Remaining Unused IP Address Range

After allocating the above subnets:

- **Remaining Network:** 192.168.100.224/27
- **Network Address:** 192.168.100.224
- **Valid Host Range:** 192.168.100.225 – 192.168.100.254
- **Broadcast Address:** 192.168.100.255
- **Usable Hosts:** 30

### Final Answer (Tabular Form)

| Department                       | Subnet     | Network Address        | Broadcast Address      | Valid Host Range                         | Usable Hosts |
|----------------------------------|------------|------------------------|------------------------|------------------------------------------|--------------|
| School of Computing              | /25        | 192.168.100.0          | 192.168.100.127        | 192.168.100.1 – 192.168.100.126          | 126          |
| School of Electronics            | /26        | 192.168.100.128        | 192.168.100.191        | 192.168.100.129 – 192.168.100.190        | 62           |
| School of Mechanical Engineering | /27        | 192.168.100.192        | 192.168.100.223        | 192.168.100.193 – 192.168.100.222        | 30           |
| <b>Unused IP Block</b>           | <b>/27</b> | <b>192.168.100.224</b> | <b>192.168.100.255</b> | <b>192.168.100.225 – 192.168.100.254</b> | <b>30</b>    |

### Summary

- **School of Computing:** Network = **192.168.100.0/25**, Broadcast = **192.168.100.127**, Hosts = **126**
- **School of Electronics:** Network = **192.168.100.128/26**, Broadcast = **192.168.100.191**, Hosts = **62**
- **School of Mechanical Engineering:** Network = **192.168.100.192/27**, Broadcast = **192.168.100.223**, Hosts = **30**
- **Remaining Unused Block:** **192.168.100.224/27** with **30 usable host addresses**.

## 2. Given

- **Allocated IP Block: 10.10.0.0/16**
- **Software Development: /24**
- **Quality Assurance: /25**
- **Human Resources: /26**
- **Executive Management: /27**

### Subnet Masks

- **/24 = 255.255.255.0**
- **/25 = 255.255.255.128**
- **/26 = 255.255.255.192**
- **/27 = 255.255.255.224**

### Step 1: Allocate the Subnets

#### 1. Software Development (/24)

- **Network Address:** 10.10.0.0
- **Subnet Mask:** 255.255.255.0
- **Broadcast Address:** 10.10.0.255
- **Valid Host Range:** 10.10.0.1 – 10.10.0.254
- **Usable Hosts:** 254

#### 2. Quality Assurance (/25)

- **Network Address:** 10.10.1.0
- **Subnet Mask:** 255.255.255.128
- **Broadcast Address:** 10.10.1.127
- **Valid Host Range:** 10.10.1.1 – 10.10.1.126
- **Usable Hosts:** 126

#### 3. Human Resources (/26)

- **Network Address:** 10.10.1.128
- **Subnet Mask:** 255.255.255.192
- **Broadcast Address:** 10.10.1.191
- **Valid Host Range:** 10.10.1.129 – 10.10.1.190
- **Usable Hosts:** 62

#### 4. Executive Management (/27)

- **Network Address:** 10.10.1.192

- **Subnet Mask:** 255.255.255.224
- **Broadcast Address:** 10.10.1.223
- **Valid Host Range:** 10.10.1.193 – 10.10.1.222
- **Usable Hosts:** 30

## Step 2: Remaining Unused IP Address Range

After allocating the above subnets, the remaining addresses in **10.10.1.x** are:

- **Unused Block:** 10.10.1.224/27
- **Network Address:** 10.10.1.224
- **Valid Host Range:** 10.10.1.225 – 10.10.1.254
- **Broadcast Address:** 10.10.1.255
- **Usable Hosts:** 30

Beyond that, since the original allocation is **10.10.0.0/16**, all addresses from **10.10.2.0** to **10.10.255.255** also remain unused.

## Final Answer (Table)

| Department           | Subnet | Network Address | Broadcast Address | Valid Host Range          | Usable Hosts |
|----------------------|--------|-----------------|-------------------|---------------------------|--------------|
| Software Development | /24    | 10.10.0.0       | 10.10.0.255       | 10.10.0.1 – 10.10.0.254   | 254          |
| Quality Assurance    | /25    | 10.10.1.0       | 10.10.1.127       | 10.10.1.1 – 10.10.1.126   | 126          |
| Human Resources      | /26    | 10.10.1.128     | 10.10.1.191       | 10.10.1.129 – 10.10.1.190 | 62           |
| Executive Management | /27    | 10.10.1.192     | 10.10.1.223       | 10.10.1.193 – 10.10.1.222 | 30           |

## Remaining Unused IP Range

- **Within 10.10.1.x:** 10.10.1.224 – 10.10.1.255 (Subnet: 10.10.1.224/27)
- **Remaining part of the /16 block:** 10.10.2.0 – 10.10.255.255

This is the correct VLSM allocation for the **10.10.0.0/16** network.

### 3. Given

- **IPv6 Address:** 2001:0db8:abcd:0000:0000:0000:0000/64

#### 1. Convert the IPv6 Address into Compressed Notation

Rules:

- Remove leading zeros in each block.
- Replace consecutive blocks of 0000 with :: (only once).

**Compressed IPv6 Address:**

**2001:db8:abcd::/64**

#### 2. Expand the Compressed Address Back into Full Hexadecimal Representation

Expand :: back to the required number of 0000 blocks.

**Expanded IPv6 Address:**

**2001:0db8:abcd:0000:0000:0000:0000/64**

#### 3. Identify the Network Prefix (/64)

With a /64 prefix:

- The **first 64 bits** represent the **network prefix**.
- The **last 64 bits** represent the **interface identifier (host portion)**.

**Network Prefix:**

**2001:0db8:abcd:0000::/64**

or simply

**2001:db8:abcd:0::/64**

#### 4. Calculate the Total Number of Host Addresses Supported Within the Subnet

A /64 prefix leaves:

- **$128 - 64 = 64$  host bits**

Total host addresses:

[  
 $2^{64} = 18,446,744,073,709,551,616$   
]

**Total Host Addresses = 18,446,744,073,709,551,616**

#### **5. Total Number of Possible Host Addresses Available Within This Network**

Since IPv6 does **not reserve network or broadcast addresses**, **all ( $2^{64}$ )** addresses are usable.

**Possible Host Addresses = 18,446,744,073,709,551,616**

#### **Final Answer (Table)**

| Task                                     | Answer                                                  |
|------------------------------------------|---------------------------------------------------------|
| <b>1. Compressed IPv6 Address</b>        | <b>2001:db8:abcd::/64</b>                               |
| <b>2. Expanded IPv6 Address</b>          | <b>2001:0db8:abcd:0000:0000:0000:0000/64</b>            |
| <b>3. Network Prefix (/64)</b>           | <b>2001:db8:abcd:0::/64</b>                             |
| <b>4. Total Host Addresses Supported</b> | <b><math>2^{64} = 18,446,744,073,709,551,616</math></b> |
| <b>5. Total Possible Host Addresses</b>  | <b>18,446,744,073,709,551,616 (<math>2^{64}</math>)</b> |

**Note:** In IPv6, unlike IPv4, there is **no broadcast address**, and a /64 subnet provides  **$2^{64}$**  possible interface addresses.

#### 4. Given

##### Routing Table

| Destination Network | Subnet Mask   |
|---------------------|---------------|
| 192.168.1.0/24      | 255.255.255.0 |
| 192.168.2.0/24      | 255.255.255.0 |
| 192.168.3.0/24      | 255.255.255.0 |

**Destination IP Address: 192.168.2.45**

##### 1. Determine the Destination Subnet

A /24 subnet means the first three octets represent the network.

- Destination IP: **192.168.2.45**
- Network Address: **192.168.2.0/24**

**Answer:** The packet belongs to the **192.168.2.0/24** subnet.

##### 2. Matching Routing Table Entry

The router checks its routing table using the **Longest Prefix Match**.

The matching entry is:

**192.168.2.0/24**

##### 3. Next-Hop Network/Interface

Since the destination belongs to **192.168.2.0/24**, the router forwards the packet through the **interface or next hop associated with the 192.168.2.0/24 route**.

**Answer:**

- **Next-hop Network: 192.168.2.0/24**
- **Outgoing Interface:** The interface configured for the **192.168.2.0/24** network (for example, **Interface G0/1**, depending on the router configuration).

**Note:** The exact interface name (such as G0/1 or Fa0/0) cannot be determined because it is not provided in the question.

##### 4. Default Route

If the destination IP does **not** match any routing table entry, the router forwards the packet using the **default route**.

**Default Route:**

**0.0.0.0/0**

This route sends the packet to the default gateway (next-hop router).

**Final Answer (Table)**

| Task                                   | Answer                                                                                |
|----------------------------------------|---------------------------------------------------------------------------------------|
| <b>1. Destination Subnet</b>           | <b>192.168.2.0/24</b>                                                                 |
| <b>2. Matching Routing Table Entry</b> | <b>192.168.2.0/24</b>                                                                 |
| <b>3. Next-Hop Network/Interface</b>   | <b>192.168.2.0/24</b> (through the interface configured for this network, e.g., G0/1) |
| <b>4. Default Route</b>                | <b>0.0.0.0/0</b> (used when no route matches)                                         |

This answer is suitable for assignment and exam purposes.

## **5. Given**

- **Administration LAN: 192.168.10.0/26**
- **Finance LAN: 192.168.10.64/26**

### **Router Interfaces (Gateways):**

- **Administration Gateway: 192.168.10.1**
- **Finance Gateway: 192.168.10.65**

**Subnet Mask (/26): 255.255.255.192**

## **1. Determine the Network Address and Broadcast Address**

### **Administration LAN**

- **Network Address: 192.168.10.0**
- **Broadcast Address: 192.168.10.63**

### **Finance LAN**

- **Network Address: 192.168.10.64**
- **Broadcast Address: 192.168.10.127**

## **2. Valid Host IP Address Range**

### **Administration LAN**

- **Host Range: 192.168.10.1 – 192.168.10.62**
- **Usable Hosts: 62**

### **Finance LAN**

- **Host Range: 192.168.10.65 – 192.168.10.126**
- **Usable Hosts: 62**

## **3. Suitable IP Addresses for Three Computers**

### **Administration LAN**

| Computer | IP Address   |
|----------|--------------|
| PC1      | 192.168.10.2 |
| PC2      | 192.168.10.3 |
| PC3      | 192.168.10.4 |

(Gateway: 192.168.10.1)

### Finance LAN

| Computer | IP Address    |
|----------|---------------|
| PC1      | 192.168.10.66 |
| PC2      | 192.168.10.67 |
| PC3      | 192.168.10.68 |

(Gateway: 192.168.10.65)

## 4. Default Gateway for Each LAN

| LAN                | Default Gateway |
|--------------------|-----------------|
| Administration LAN | 192.168.10.1    |
| Finance LAN        | 192.168.10.65   |

## Final Answer (Table)

| Department         | Network Address  | Broadcast Address | Valid Host Range               | Example Host IPs                                  | Default Gateway |
|--------------------|------------------|-------------------|--------------------------------|---------------------------------------------------|-----------------|
| Administration LAN | 192.168.10.0/26  | 192.168.10.63     | 192.168.10.1 – 192.168.10.62   | 192.168.10.2,<br>192.168.10.3,<br>192.168.10.4    | 192.168.10.1    |
| Finance LAN        | 192.168.10.64/26 | 192.168.10.127    | 192.168.10.65 – 192.168.10.126 | 192.168.10.66,<br>192.168.10.67,<br>192.168.10.68 | 192.168.10.65   |

This is the correct subnetting and addressing plan for the two LANs and is suitable for assignment or exam submission.